Scott Nguyen

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EDUCATION

Doctor of Philosophy in Mechanical and Aerospace Engineering **Summer 2030 (Expected)**
University of California San Diego | San Diego State University

Master of Science in Electrical Engineering | Space Systems Engineering **Summer 2026**
University of New Mexico GPA: 3.93/4.00

Master of Science in Aerospace Engineering **Fall 2024**
University of Illinois Urbana-Champaign

Bachelor of Science in Aerospace Engineering **Spring 2022**
Iowa State University

TEACHING & OUTREACH EXPERIENCE

Youth Development Professional **October 2024 – January 2025**
Boys & Girls Club
- Designed and delivered an after-school computer literacy curriculum for grades 3–6.
- Taught introductory coding through Scratch and beginner Python using project-based activities.
- Used quick assessments and differentiated support to improve student engagement and learning.

Math Teacher (Algebra I, Algebra II, Precalculus, Calculus) **October 2024 – January 2025**
North Star School
- Developed unit plans, assessments, and problem sets aligned with course objectives.
- Differentiated instruction and provided standards-based feedback.
- Connected mathematics to real-world STEM and engineering applications.

Research Assistant, AE 298 National Defense Education Program **January 2023 – May 2024**
Aerospace Engineering, University of Illinois Urbana-Champaign
- Taught an introductory rocketry course for undergraduate engineering students.
- Collected and analyzed program data to evaluate educational impact.
- Redesigned course structure to improve retention and learning outcomes.

Research Assistant, Grants for Advancement of Teaching in Engineering (GATE) **November 2023 – May 2024**
Mechanical Engineering, University of Illinois Urbana-Champaign
- Built a comprehensive engineering final project plan using interdisciplinary pedagogy.
- Conducted engineering education research and contributed to conference publications.

Teaching Assistant, ASTR 122/210 **August 2023 – May 2024**
Astronomy, University of Illinois Urbana-Champaign
- Led weekly discussions and created exam reviews for 150+ students.
- Produced personalized study materials and graded assignments and examinations.

Graduate Career Peer **August 2022 – May 2024**
Engineering Career Services, University of Illinois Urbana-Champaign
- Advised students on career development, resumes, and interview preparation.
- Conducted mock interviews and provided actionable feedback.

Graduate Assistant, National Defense Education Program **January 2023 – May 2023**
Aerospace Engineering, University of Illinois Urbana-Champaign
- Built partnerships with local K–12 schools for rocketry and avionics initiatives.
- Mentored educators and students and supported outreach assessment efforts.

Private Tutor — SAT/ACT Test Prep & Math/Science **January 2020 – May 2022**
Independent
- Delivered individualized SAT/ACT preparation and mathematics/science tutoring.
- Created personalized study plans and tracked student progress through assessments.

Calculus, Physics, and Differential Equations Tutor **January 2019 – May 2022**
Academic Success Center, Iowa State University
- Delivered tailored lessons for diverse learners in mathematics, physics, and engineering.
- Helped students improve course performance and conceptual understanding.

AEROSPACE INDUSTRY EXPERIENCE

Guidance, Navigation & Controls Engineer Intern **May 2026 – Present**
Air Force Research Laboratory (AFRL)
- Implemented and evaluated advanced pose-estimation algorithms for robotics and autonomous systems.
- Conducted Monte Carlo analysis and simulation-based validation using ROS2 and Gazebo.
- Supported autonomous navigation research for aerial and ground vehicles.

Guidance, Navigation & Controls Engineer Intern **January 2026 – May 2026**
NASA Johnson Space Center
- Developed Guidance, Navigation, and Controls flight software within NASA's core Flight System.
- Supported autonomous vehicle simulations using PX4, MAVSDK, and Trick.
- Worked with camera, IMU, and spacecraft software integration efforts.

Student Co-Op **September 2025 – January 2026**
MIT Lincoln Laboratory, Electronics for Contested Space Group
- Developed satellite tracking and orbit propagation tools for Space Domain Awareness applications.
- Modeled space-object visibility and sensor performance for observation planning.

Guidance, Navigation & Controls Engineer Intern **May 2025 – August 2025**
Blue Canyon Technologies
- Executed testing and analysis of spacecraft sensors and attitude-control hardware.
- Worked with IMUs, star trackers, reaction wheels, sun sensors, and spacecraft avionics.

Guidance, Navigation & Controls Engineer Intern **January 2025 – April 2025**
Blue Origin
- Designed and evaluated advanced control algorithms for aerospace applications.
- Performed simulation-based testing and flight software integration activities.

Guidance, Navigation & Controls Engineer Intern **May 2024 – August 2024**
Varda Space Industries
- Performed spacecraft navigation trade studies involving estimation and flight-safety analysis.
- Developed simulation tools supporting reentry navigation architecture evaluation.

Guidance, Navigation & Controls Engineer Intern **May 2023 – August 2023**
Space Dynamics Laboratory
- Developed orbit-determination algorithms and space-surveillance analysis tools.
- Conducted Monte Carlo studies and implemented initial orbit determination methods.